AGRIVIVE / LOKTAK-ANCHOR

MANIPUR / LOKTAK-INSPIRED / ROOT SUPPORT AND MOORING FRAME

SERIES 15 / 28

A3 / PRELIMINARY / ALL DIMENSIONS mm



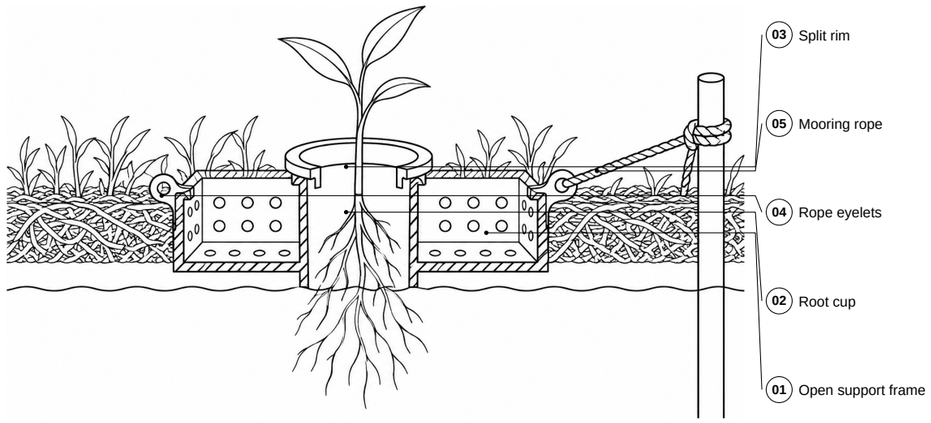
STATE CONTEXT / INDIGENOUS INSPIRATION

Vegetation-based floating cultivation is documented around Loktak Lake in Manipur. LOKTAK-ANCHOR proposes a root cup and tie points attached to an existing plant-bed mat; the frame itself provides no claimed buoyancy.

Practice source: CEEW: State of floating farming in India | Outline: geoBoundaries / schematic

01 INSTALLED SOIL / WATER SECTION

NOMINAL ENVELOPE / mm: 120 square anchor frame x 50 H

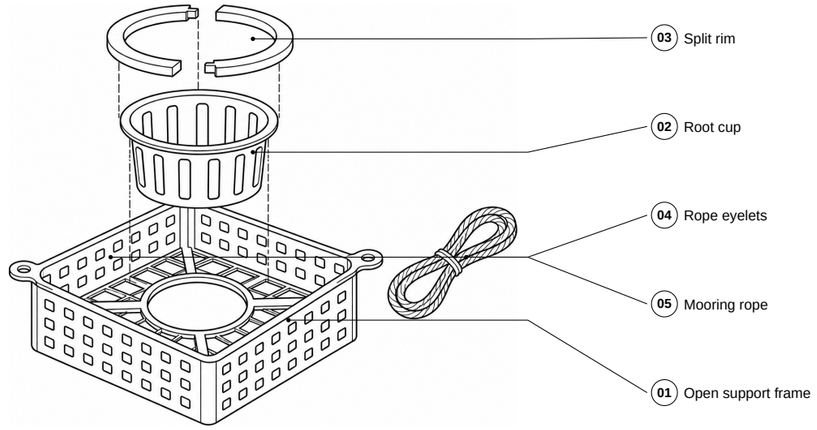


DESIGN / OPERATION NOTES

Fit the frame into an existing vegetation-bed structure.
Seat the root cup and rim with the root passage open below.
Tie to an appropriate existing support; assess the full bed and mooring system.

02 EXPLODED ASSEMBLY

ASSEMBLED ENVELOPE / mm
120 square anchor frame x 50 H

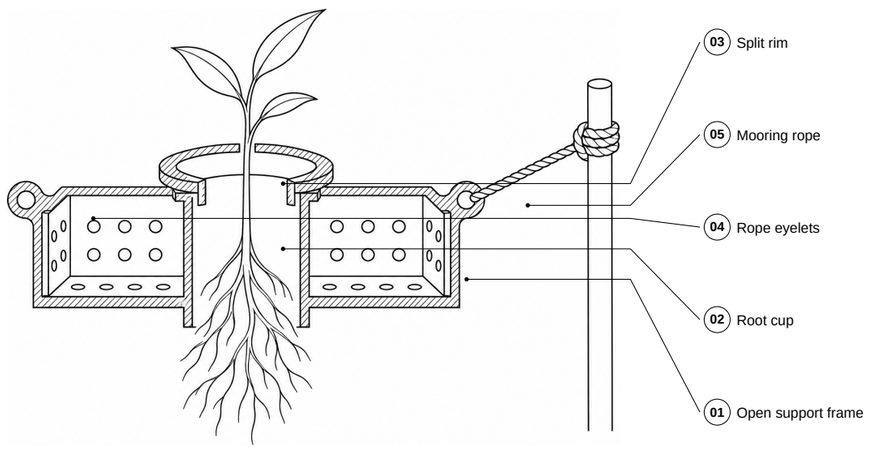


DESIGN / OPERATION NOTES

Components separated for service and assembly.
Connections and mating fits require CAD.

04 INTERNAL SECTION

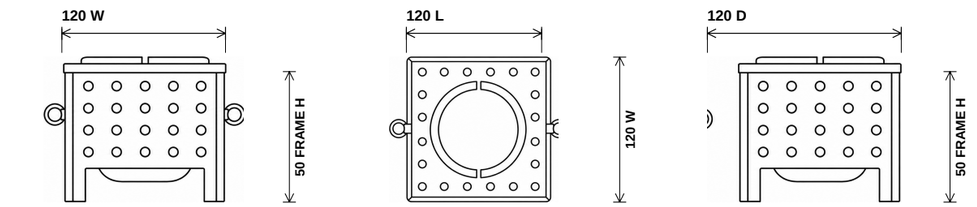
SECTION BASIS / mm
50 planting-cup ID x 50 clear H; mooring eye sizes TBD



DESIGN / OPERATION NOTES

Seat the root cup and rim with the root passage open below.
Tie to an appropriate existing support; assess the full bed and mooring system.

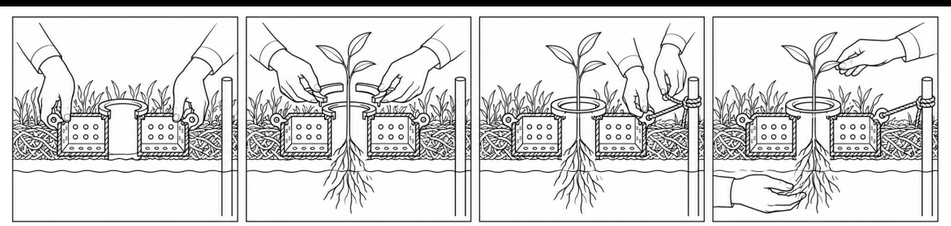
03 ORTHOGRAPHIC STUDIES / mm



FRONT / FRAME **TOP / FRAME** **SIDE / FRAME**

Frame 120 square x 50 H. Planting pocket DIA 50 x 50 clear depth; pocket volume is not flotation capacity.

05 INSTALLATION / OPERATING SEQUENCE



1 POSITION FRAME **2 FIT ROOT CUP** **3 TIE MOORING** **4 INSPECT ROOTS**

Prototype fit, access and operation require trials.

DESIGN / INTERFACE SCHEDULE

01 NOMINAL GEOMETRY

Frame L x W x H
120 x 120 x 50

Cup ID / clear depth
50 / 50

Eyelet / rope detail
TBD

02 GEOMETRIC CHECK

GROSS PLANTING POCKET

98.2 mL

pi/4 x 50^2 x 50 / 1000
Substrate pocket; not buoyancy or mooring-load capacity

03 PARTS / QTY

01 Open support frame	**1**
02 Root cup	**1**
03 Split rim	**1 set**
04 Rope eyelets	**2 integral**
05 Mooring rope	**As needed**

Frame/cup/rim: candidate polymer. Mooring rope grade, diameter and termination TBD.

04 INTERFACES / CHECKS

Test cup retention, root clearance and service access.

Check rope abrasion and mooring loads.

Assess existing bed buoyancy independently.

05 DATUMS / DRAWING BASIS

A: frame upper plane. B: cup rim. Keep the lower root passage open.
PRELIMINARY / VALIDATION PENDING

DESIGN DEVELOPMENT RECORD

01 / GEOMETRY
Nominal body sizes are proposed. Generated views are illustrative; small features and interfaces may vary.

02 / BEFORE FABRICATION
Develop one CAD model, assign wall thicknesses, tolerances and mating fits. Check print orientation and service access.

03 / TRIAL RECORD
Measure dimensions and retained volume or opening area. Record leakage, clogging and soil anchorage under observed conditions.

No discharge, treatment or durability performance is established.

DRAWING / REVISION	TITLE	STATUS	REVISION DATE	FORMAT / SCALE / UNITS	COLLECTION SHEET
AGR-15 / P01	LOKTAK-ANCHOR	PRELIMINARY CONCEPT / CAD & TRIALS PENDING	10 NOV 2026	A3 / NTS / mm	15 / 28